

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION
SSC PART-I (CLASS 9) PRACTICE EXAMINATION
MATHEMATICS PAPER 19

Syllabus: Unit 11 Basic Statistics (24 marks) + Unit 10 Practical Geometry (15 marks) + Unit 6, sections 6.1 and 6.2 only (11 marks)

Note: Unit 6 (6.1 Measurement of an Angle, 6.2 Sector of Circle) abhi class mein nahin parhaya gaya. Ye hissa sirf 11 marks ka hai, is liye pareshan na hon.

Total Marks: 50

Time Allowed: 2 hours 15 minutes

Version A

HIDAYAAT (Instructions)

1. **Saaray sawal** hal karnay hain. Is paper mein koi bhi **optional sawal nahin** hai.
2. Jawab apni notebook mein likhein. Har sawal ka number **bilkul waisa hi** likhein jaisa paper par hai, aur tarteeb se hal karein.
3. Section A mein sawal ka number likh kar sirf apnay option ka **letter** likhein, jaise **1. (b)**. Check kar lein ke jo letter aap ne likha hai, wohi option aap ne choose kiya hai.
4. Har calculation mein pehlay **formula** likhein, phir working. **Calculator use kar sakti hain**. Har jawab ke saath **unit** zaroor likhein (cm, cm², radians waghera).
5. **Sawal 9 ke liye geometry box zaroori hai** (ruler, compass, protractor). Drawing **pencil** se karein, baqi paper pen se.
6. Construction wale sawal mein **arcs aur construction lines mitayein nahin**, unhi par marks hain.
7. Har lambay sawal ke aakhir mein **ek mukammal jumla** likhein jo sawal ka jawab de.

SECTION A: MULTIPLE CHOICE QUESTIONS (10 × 1 = 10 MARKS)

#	Question	A	B	C	D	Answer
1	How many seconds are there in one minute of an angle?	6	10	60	360	(A)(B)(C)(D)
2	1 radian is approximately equal to:	57°17'45"	45°	60°	90°	(A)(B)(C)(D)
3	The area of a sector of a circle of radius r and central angle θ radians is:	$A = r\theta$	$A = 2\pi r$	$A = \pi r^2$	$A = \frac{1}{2}r^2\theta$	(A)(B)(C)(D)
4	The point where the three angle bisectors of a triangle meet is called the:	centroid	in-centre	orthocentre	circumcentre	(A)(B)(C)(D)
5	The right bisectors of a right-angled triangle meet at the:	vertex of the right angle	centroid of the triangle	mid point of the hypotenuse	point outside the triangle	(A)(B)(C)(D)
6	The altitudes of an obtuse-angled triangle intersect:	inside the triangle	at the mid point of the longest side	at the vertex of the obtuse angle	outside the triangle	(A)(B)(C)(D)
7	The class mark (mid value) of the class 5 to 15 is:	5	10	15	20	(A)(B)(C)(D)
8	The most frequent value of a data is called its:	mean	median	mode	class mark	(A)(B)(C)(D)
9	The probability of an impossible event is:	0	$\frac{1}{2}$	1	2	(A)(B)(C)(D)
10	A fair coin is tossed 400 times. The expected number of heads is:	100	200	40	400	(A)(B)(C)(D)

SECTION B: SHORT ANSWER QUESTIONS (6 × 4 = 24 MARKS)

Chhe ke chhe sawal hal karein. Har sawal 4 marks ka hai.

2.	Angle conversions. (i) Convert 75° into radians. Give your answer in terms of π . (2) (ii) Convert $5\pi/6$ radians into degrees. (2)	4
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3.	A sector of a circle has radius $r = 6 \text{ cm}$ and central angle $\theta = \pi/3$ radians. Taking $\pi = 3.14$, find: (i) the length of the arc of the sector. (2) (ii) the area of the sector. (2) Write the formula first in each part, and give the unit with every answer.	4
4.	A frequency distribution has the classes 5 to 14, 15 to 24, 25 to 34, 35 to 44 . For the class 15 to 24 , write: (i) its lower class limit and its upper class limit. (1) (ii) its class boundaries. (1) (iii) its class mark. (1) (iv) the size of the class interval, h . (1)	4
5.	Find the mean , the median and the mode of the following data: 4, 6, 9, 6, 10, 7, 6, 12 (i) mean (2) (ii) median (1) (iii) mode (1) For the median, first write the data in ascending order.	4
6.	A bag contains 5 red, 3 green and 2 blue marbles of the same size. One marble is drawn at random. Find the probability that it is: (i) green (1) (ii) blue (1) (iii) not red (1) (iv) A fair die is rolled 300 times. How many times is a 4 expected to come up? (1)	4
7.	Lines inside a triangle. (i) Define the altitude of a triangle. (1) (ii) Define the median of a triangle. (1) (iii) What is the point called where the three altitudes of a triangle meet? (1) (iv) In what ratio does the centroid divide each median? (1)	4

SECTION C: LONG ANSWER QUESTIONS (2 × 8 = 16 MARKS)

Dono sawal hal karein. Har sawal **8 marks** ka hai.

8.	Marks obtained by 30 students are given below. <table><tr><th>Marks</th><td>1 to 10</td><td>11 to 20</td><td>21 to 30</td><td>31 to 40</td><td>41 to 50</td></tr><tr><th>No. of students (<i>f</i>)</th><td>4</td><td>6</td><td>10</td><td>6</td><td>4</td></tr></table> (i) Copy the table into your notebook and add columns for class boundaries, class marks (<i>x</i>), <i>fx</i> and cumulative frequency. (2) (ii) Find the arithmetic mean using $\bar{x} = \Sigma fx / \Sigma f$. (2) (iii) Find the median using $\tilde{x} = l + (h/f)(n/2 - c)$. State the median group first. (2) (iv) Find the mode using $\text{Mode} = l + [(f_m - f_1) / (2f_m - f_1 - f_2)] \times h$. (1) (v) Write one sentence stating what your three answers tell you about this data. (1)	Marks	1 to 10	11 to 20	21 to 30	31 to 40	41 to 50	No. of students (<i>f</i>)	4	6	10	6	4	8
Marks	1 to 10	11 to 20	21 to 30	31 to 40	41 to 50									
No. of students (<i>f</i>)	4	6	10	6	4									
9.	Construction. Using only your ruler and compasses, construct a triangle ABC in which AB = 6 cm, BC = 5 cm and CA = 4.5 cm. (i) Draw the triangle and write the steps of construction. (3) (ii) Draw the bisectors of ∠A, ∠B and ∠C, and mark the point O where they meet. (3) (iii) Are the three angle bisectors concurrent? Write one sentence naming the point O and saying where it lies. (2) Apne arcs aur construction lines mat mitayein.	8												

END OF QUESTION PAPER

Units covered: Unit 11 Basic Statistics (MCQs 7 to 10, Questions 4, 5, 6 and 8) | Unit 10 Practical Geometry (MCQs 4 to 6, Questions 7 and 9) | Unit 6 Trigonometry and Bearing, sections 6.1 and 6.2 only (MCQs 1 to 3, Questions 2 and 3).

Student's Name: _____

Date: _____

Marks Obtained: _____ / 50